# Student Assignment Brief

**This document is intended for Coventry University Group students for their own use in completing their assessed work for this module. It must not be passed to third parties or posted on any website. If you require this document in an alternative format, please contact your Module Leader.**

## Contents:


- Assignment Information
- Assignment Task
- Marking and Feedback
- Assessed Module Learning Outcomes
- Assignment Support and Academic Integrity
- Assessment Marking Criteria


The work you submit for this assignment must be your own independent work, or in the case of a group assignment your own groups' work. More information is available in the 'Assignment Task' section of this assignment brief.

## Assignment Information

**Module Name:** Propulsion Systems and Aerodynamics

**Module Code:** 7042ENG

**Assignment Title:** Normal Course work

**Assignment Due:** 20th of April at 18:00

If you do not pass this assessment, you may have an opportunity to resit it. **If you do need to resist, you will be asked to use the feedback provided to revise your original submission, so that it meets the pass requirements for the module.** You must clearly indicate the changes you have made in the new submission. Please check your SOLAR results and the submission links on your Aula module page to see when the rest is due.

**Assignment Credit:** 30 credits

**Word Count (or equivalent):** **4000 words total**, (**Task 1= 1500 words** and **Task 2=2500 words**) +/- 10%

**Assignment Type:**

**Percentage Grade** (Applied Core Assessment). You will be provided with an overall grade between 0% and 100%. You have one opportunity to pass the assignment at or above 40%.

# Assignment Task

As part of your assessment, you are required to submit a single engineering report of 4,000 words ±10%, structured with numbered sections for each task. Figures must be captioned and referenced in the text. Equations numbered. APA references. Figures, tables, and references excluded from word count.

### Task 1 (Analytical)

Consider you work as an aircraft designer looking into supersonic research aircraft (Bristol 188) designed to achieve a speed of Mach 3 (measure in front of the shock wave) at 13,000 m in altitude. As for achieving Mach 2.5 at 11,000 it is required to produce 89 kN of thrust.

You have been tasked to produce the preliminary design analysis of a selected engine to be mounted in the aircraft. For that and looking into the engine properties described below, complete the follow task:

- Calculate the air properties inside the shock wave for the following speeds and deflection angle of the shockwave (Table 1).
- Calculate the engine theoretical thrust given the engines parameters (Table 2).

You also need to briefly discuss the following:

- Justify any assumptions made during the calculations.
- Critically evaluate the accuracy and limitations of your results.
- Discuss how component efficiencies and ambient conditions influence performance.
- Propose how to increase performance to reach design requirements.

*Table 1: Flow parameters*

| Upstream flow parameters | | |
|---|---|---|
| **Velocity** $(\boldsymbol{MACH})$ | **Altitude** $(\boldsymbol{m})$ | **deflection angle** $(\boldsymbol{\theta})$ |
| 1.6 | 9,000 | 13.05° |
| 2.5 | 11,000 | 13.05° |
| 3 | 13,000 | 13.05° |

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*Table 2.- Engine parameters*

| | | |
|---|---|---|
| **Compressor pressure ratio** | $\pi_c$ | 6.4 |
| **Intake efficiency** | $\eta_d$ | 0.92 |
| **Compressor efficiency** | $\eta_c$ | 0.88 |
| **Burner efficiency** | $\eta_b$ | 0.99 |
| **Turbine efficiency** | $\eta_t$ | 0.96 |
| **Nozzle efficiency** | $\eta_n$ | 0.9 |
| **Burner Pressure loss** | $\Delta P_b$ | 0.95 |
| **Maximum Temperature** | $T_{max}\ (°K)$ | 1073 |
| **Fuel heating value** | $Q_R\ (kJ/kg)$ | 48,000 |
| **Inlet mass flow rate** | $\dot{m}_a\ (kg/s)$ | 56 |

Finally, you are allowed to create a computer code to solve this, but it need to be explained, and the code attached in the appendix with the required comments.

**Task 2 (CFD)**

You have been recruited as a CFD engineer at a major aero-engine manufacturer investigating the performance of a high-pressure (HP) turbine nozzle guide vane (NGV) row. HP NGVs must turn a supersonic, highly swirling gas stream whilst minimising aerodynamic losses that directly reduce specific fuel consumption (SFC). This task mirrors that industrial workflow: you will set up and run ANSYS Fluent simulations of a 2D NGV passage, analyse the transonic flow field, propose a geometry improvement, and critically reflect on your findings.

Define a single-passage 2D NGV geometry in ANSYS DesignModeler or SpaceClaim within the following parametric ranges:

| Parameter | Specification | Notes |
|---|---|---|
| Chord length, C | 60 mm | Axial chord Ca ≈ 0.6C |
| Pitch-to-chord ratio, s/C | 1.5 | Target Zweifel (Z_w) ≈ 1.7-3.7 |
| Stagger angle | 60° – 70° | From axial direction |
| Inlet flow angle $\beta_1$ | 0° (axial) | Typical for NGV row |
| Outlet flow angle $\beta_2$ | 61° – 72° | High turning for HP NGV |
| Leading edge radius | 2–5% of C | Avoid sharp LE |
| Trailing edge thickness | ≤1.5% of C | Minimise base drag |
| Blade profile family | C4, DCA, or bespoke cambered aerofoil | Must suit transonic conditions |

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### Task 2.1 – Geometry and Meshing

Create the 2D NGV geometry in DesignModeler or SpaceClaim using the specification in above, documenting and justifying all chosen parameter values against the Zweifel criterion. Set the computational domain with the inlet ≥1.5C upstream and outlet ≥2C downstream, with translational periodic pitch-wise boundaries.

Generate a structured or hybrid mesh targeting $y^+ < 1$ on all blade surfaces (required for k-ω SST low-Re mode). Calculate and document the required first-cell height analytically. Apply refinement near the leading edge, trailing edge, and wake. Report minimum orthogonal quality (≥0.1) and maximum skewness (≤0.85).

Keep in mind that for conventional turbine blade design, a target of Z_w ≈ 0.8–0.9 is typically cited in the literature (Zweifel, 1945) as the optimum range that balances blade loading against aerodynamic loss. Whilst the target Z_w is higher than the classical optimum, it remains a physically meaningful and internally consistent design specification for this design task. Critically reflect and discuss what this implies for the blade loading relative to a real engine design referring to the conditions at which a target of Z_w ≈ 0.8–0.9 was dervived.

### Task 2.2 - Transonic Nominal Operating Condition

Configure ANSYS Fluent for the transonic design point using the boundary conditions below. Justify your choice of density-based solver and Roe-FDS scheme.

| Parameter | Value / Setting |
|---|---|
| Inlet total pressure, $P_{01}$ | 400 kPa |
| Inlet total temperature, $T_{01}$ | 1,400 K |
| Inlet turbulence intensity | 5%; length scale = 2% chord |
| Exit static pressure | Set to achieve isentropic exit Mach ≈ 1.0–1.2; calculate and justify |
| Working fluid | Ideal gas (air); γ = 1.33, $Cp$ = 1,148 J/kg·K |
| Turbulence model | $k-\omega$ SST (low-Re correction enabled) |
| Solver | Density-based, implicit; Roe-FDS; second-order upwind |
| Convergence | Residuals < $10^{-6}$ AND stabilised $\omega, \beta_2$, and mass flow rate |

Present Mach number contours, surface $Cp$ distribution, numerical Schlieren (density gradient magnitude), and turbulent kinetic energy contours. Calculate and tabulate: total pressure loss coefficient $\omega$ (mass-averaged), exit flow angle $\beta_2$ and deviation from design intent, isentropic velocity coefficient $Cv$, and Zweifel coefficient from surface pressure

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integration. Identify and quantify all major loss mechanisms including trailing edge loss, shock system, shock–boundary layer interaction (SBLI), and wake mixing loss.

### Task 2.3 - Performance-Driven Design Optimisation

* * Please note that your modified blade must not reduce exit flow angle $\beta_2$ by more than 2° from the baseline.

Based on your loss analysis from Task 2, propose and implement one geometry modification to reduce aerodynamic losses. The modification must be directly motivated by a specific, quantified loss mechanism and physically justifiable using turbomachinery theory. Suggested approaches include: trailing edge wedge angle/cutback, suction surface curvature redistribution to reposition the shock, or blade loading redistribution (fore- vs. aft-loading) to reduce SBLI.

Write a short design intent paragraph (~150 words) stating the targeted loss mechanism, the aerodynamic reasoning, the specific geometric change, and the expected outcome. Re-mesh the modified geometry with comparable $y^+$ and cell count, run the transonic condition, and present side-by-side visualisations (Mach contours, numerical Schlieren, surface $Cp$) alongside a quantitative performance table showing percentage change in $\omega, \beta_2$, and $Cv$. Conclude with a critical evaluation: did the modification work as expected, what are the limitations of this 2D RANS approach, and what industrial optimisation methods (e.g. adjoint methods, surrogate models) would offer further capability?

### Task 2.4 - Critical Discussion

Critically evaluate the reliability of your CFD predictions: where is 2D steady RANS most and least trustworthy for this flow? Discuss the strengths and weaknesses of $k-\omega$ SST (adverse pressure gradient handling, SBLI capability, transition prediction), and estimate qualitatively how 3D effects absent from your model - tip clearance, end wall secondary flows, spanwise loading variation — would affect the total loss coefficient in a real engine.

Without running additional simulations, discuss how the NGV would behave at a subsonic idle condition (isentropic exit Mach ≈ 0.6–0.75): how would blade loading, shock structures, boundary layer state, and turning angle change? Compare your baseline loss coefficient against at least one analytical framework (e.g. Ainley-Mathieson correlation, isentropic shock loss relations, or Zweifel loading assessment) and explain any discrepancies. Finally, reflect briefly on one industrial implication — either the estimated SFC impact of your measured losses, or the effect of film cooling ejection on the aerodynamic loss budget.

Please ensure your findings are presented in accordance with the structure and requirements outlined in the table below.

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**Report section breakdown.**

The following tables outline the report structure breakdown and marks for the report. If it is recommended that your report include section headers that reflect the marks breakdown show at the end of this document.

| Section | Marks |
|---|---|
| Introduction Task 1 | 5% |
| Methodology Task 1 | 10% |
| Task 1 results, analysis, and discussion. | 20% |
| Conclusions Task 1 | 5% |
| Introduction Task 2 | 5% |
| Task 2 Methodology (Geometry, Meshing & Independence Study) | 10% |
| Task 2 Transonic nominal condition analysis | 5% |
| Task 2 Performance-Driven design optimisation analysis | 5% |
| Task 2 Discussion. | 10% |
| Task 2 Conclusions | 5% |
| Referencing | 10% |
| Appendix/Plots | 10% |

## Artificial Intelligence notice

AI use is **allowed to assist in the development of the assessment** in line with the student guidance. However, is **not allowed in the design of components, perform the simulations, write the computer code or any part of the written report**.

## Submission Instructions:

Submission is done through AULA page under the assignment section.

- Formatting of **submission should be in Microsoft Word only** and not PDF.
- ANSYS Case Files (ZIP) containing Geometry, mesh, case and data files for all independence study meshes, baseline transonic case, and optimised geometry case. Include a README.txt referencing each file to the relevant report section.
- If using Artificial Intelligence (AI) tools in the development of your assignment, you must reference which tools you have used and for what purposes you have used them

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# Marking and Feedback

### How will my assignment be marked?

Your assignment will be marked by the lecturing team

### How will I receive my grades and feedback?

**Provisional marks will be released once internally moderated**

Feedback will be by the module team alongside grades release in Aula.

Your provisional marks and feedback should be available within 2 weeks (10 working days)

### What will I be marked against?

Details of the marking criteria for this task can be found at the bottom of this assignment brief.

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# Assessed Module Learning Outcomes

The Learning Outcomes for this module align to the marking criteria which can be found at the end of this brief. Ensure you understand the marking criteria to ensure successful achievement of the assessment task. The following module learning outcomes are assessed in this task:

***"Determine appropriate wing geometry for a given aircraft's requirements using appropriate 2D and 3D methods."***

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# Assignment Support and Academic Integrity

If you have any questions about this assignment please see the Student Guidance on Coursework for more information.

### Spelling, Punctuation, and Grammar:

You are expected to use effective, accurate, and appropriate language within this assessment task.

### Academic Integrity:

The work you submit must be your own, or in the case of groupwork, that of your group. All sources of information need to be acknowledged and attributed; therefore, you must provide references for all sources of information and acknowledge any tools used in the production of your work. We use detection software and make routine checks for evidence of academic misconduct.

It is your responsibility to keep a record of how your thinking has developed as you progress through to submission. Appropriate evidence could include: version controlled documents, developmental sketchbooks, or journals. This evidence can be called upon if we suspect academic misconduct.

If using Artificial Intelligence (AI) tools in the development of your assignment, you must reference which tools you have used and for what purposes you have used them. This information must be acknowledged in your final submission.

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Definitions of academic misconduct, including plagiarism, self-plagiarism, and collusion can be found on the Student Portal. All cases of suspected academic misconduct are referred for investigation, the outcomes of which can have profound consequences to your studies. For more information on academic integrity please visit the Academic and Research Integrity section of the Student Portal.

### Support for Students with Disabilities or Additional Needs:

If you have a disability, long-term health condition, specific learning difference, mental health diagnosis or symptoms and have discussed your support needs with health and wellbeing you may be able to access support that will help with your studies.

If you feel you may benefit from additional support, but have not disclosed a disability to the University, or have disclosed but are yet to discuss your support needs it is important to let us know so we can provide the right support for your circumstances. Visit the Student Portal to find out more.

### Unable to Submit on Time?

The University wants you to do your best. However, we know that sometimes events happen which mean that you cannot submit your assessment by the deadline or sit a scheduled exam. If you think this might be the case, guidance on understanding what counts as an extenuating circumstance, and how to apply is available on the Student Portal.

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## Administration of Assessment

**Module Leader Name:** Dr Martin Carrizales

**Module Leader Email:** ad9207@coventry.ac.uk

**Assignment Category:** Written

**Attempt Type:** Standard/Resit

**Component Code:** Cw1

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## Assessment Marking Criteria

Guidance for Course Teams: Assessment criteria should align to the University-level assessment criteria for the relevant level of study. These are available through the Academic Enhancement and Professional Development website. Training on assessment and feedback approaches is available through the Course Booking System.

The following is the marking criterion to be used in the marking of this assessment.

| | **Task 1 Introduction**<br><br>Weighting: 5% | **Task 1 Methodology**<br><br>Weighting: 10% | **Task 1 Results and discussion**<br><br>Weighting: 20% | **Task 1 Conclusions**<br><br>Weighting: 5% | **Task 2 Introduction**<br><br>Weighting: 5% | **Task 2 Methodology (Geometry, Meshing & Independence Study)**<br>Weighting: 10% | **Task 2 Transonic nominal condition analysis**<br><br>Weighting: 5% | **Task 2 Performance-Driven design optimisation analysis**<br><br>Weighting: 5% | **Task 2 Discussion**<br><br>Weighting: 10% | **Task 2 Conclusions**<br><br>Weighting: 5% | **Referencing**<br><br>Weighting: 10% | **Appendix/Plots**<br><br>Weighting: 10% |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 90 to 100% | Scientific Journal level of writing standards. | Novelty in the model design. | Novel results. Scientific Journal standards of results analysis and discussion. | Scientific Journal standard of concluding remarks. | Scientific Journal level of writing standards. | Novelty in the model design and mesh. | Novel results. Scientific Journal standards of results analysis. | Novel results. Scientific Journal standards of optimization methods and results analysis. | Novel results. Scientific Journal standards of results discussion. | Scientific Journal standard of concluding remarks. | Scientific Journal standard of use of referencing style. | Scientific Journal standard plots and Appendix. |
| 80 to 89% | Conference level of writing standards. | Conference level of design and meshing justification. | Conference levels of results analysis and discussion. | Conference level of conclusions. | Conference level of writing standards. | Conference level of design and meshing justification. | Conference levels of results analysis. | Conference levels of optimization methods and results analysis. | Conference levels of results discussion. | Conference level of conclusions. | Conference standard use of referencing style | Conference standards of plots and Appendix. |
| 70 to 79% | An excellent introduction. It's succinct (yet complete). The work is clearly motivated. And the aim and objectives are excellent. Excellent use of the literature to justify working. | All aspects related to the methodology are included. May include a deeper discussion of subsonic/transition/supersonic flow, engine design analysis. Excellent justification of numerical choices backed up with the literature. | The discussion is excellent, mature, and critical. It includes clear links to the flow physics and engineering implications Excellent use of the literature to strengthen the argument or discussion. | An excellent summary of the work done, and several key findings are included and discussed or contextualize. An excellent discussion of the implications of the work. | An excellent introduction. It's succinct (yet complete). The work is clearly motivated. And the aim and objectives are excellent. Excellent use of the literature to justify working. | All aspects related to the methodology are included. It included an excellent grid independence study, domain schematic, and boundary conditions are justified. May include a deeper discussion of turbulence/transition/laminar flow model selections. Excellent | The transonic results are excellent, mature, and well presented. It includes clear links to the flow physics and engineering implications Excellent use of the literature to strengthen the argument or discussion. | The optimization results are excellent, mature, and well presented. It includes clear links to the flow physics and engineering implications Excellent use of the literature to strengthen the argument or discussion. | The discussion is excellent, mature, and critical. It includes clear links to the flow physics and engineering implications Excellent use of the literature to strengthen the argument or discussion. | An excellent summary of the work done, and several key findings are included and discussed or contextualize. An excellent discussion of the implications of the work. | Excellent use of references. Used to strengthen the argument and justify deductions. Used to inform the numerical methodology and explain the flow physics (when appropriate). Excellent written communication. | Excellent quality of plots and Appendix. Almost publication standard (adheres to the format of the whole document). Perfectly labelled and selected to complement the narrative. Excellent discussion accompanies the plots. |

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| | | | | | | justification of numerical choices backed up with the literature. | | | | | | |
| 60 to 69% | Very good introduction. It's succinct (yet mostly) complete. Good motivation for the work is included. Very good use of the literature to justify the work. | Most aspects related to the methodology are presented. Included analysis of all the condition presented in the problem. Very good justification of the numerical choices backed up with the literature. | The discussion is very good mature and/or critical at times. It includes some links to the aerodynamics theories and engineering implications. Very good use of the literature to strengthen the argument or discussion. | A very good summary of the work done, and several key findings are included. Very good discussion of the implications of the work. | Very good introduction. It's succinct (yet mostly) complete. Good motivation for the work is included. Very good use of the literature to justify the work. | Most aspects related to the methodology are presented. Included grid independence study, domain schematic, and boundary conditions. Very good justification of the numerical choices backed up with the literature. | The results are very good mature and/or critical at times. It includes some links to the aerodynamics theories and engineering implications. Very good use of the literature to strengthen the argument or discussion. | The optimization methods and results are very good mature and/or critical at times. It includes some links to the aerodynamics theories and engineering implications. Very good use of the literature to strengthen the argument or discussion. | The discussion is very good mature and/or critical at times. It includes some links to the aerodynamics theories and engineering implications. Very good use of the literature to strengthen the argument or discussion. | A very good summary of the work done, and several key findings are included. Very good discussion of the implications of the work. | Very good use of references. Sometimes used to strengthen the argument and justify deductions. Sometimes used to inform the numerical methodology and explain the aerodynamic theories (at times). Very good written communicatio n. | Very good plot and Appendix quality. Short of publication standard (may not fully adhere to the format of the whole document). Plots are labeled very well and may complement the narrative. Very good discussion accompanies plots. |
| 50 to 59% | A good introduction. It's succinct (with minor omissions). Some attempts included motivating the work. Good aim and objectives. Good use of the literature to justify the work. | Some aspects related to the methodology are included. Good justification of numerical choices backed up with the literature to some extent. | The discussion is good. It may not be mature and/or critical. It includes very few links to aerodynamics theories and engineering implications. Good use of the literature to strengthen the argument or discussion occasionally. | A good summary of the work done, and a few key findings are presented. A good discussion of the implications of the work. | A good introduction. It's succinct (with minor omissions). Some attempts included motivating the work. Good aim and objectives. Good use of the literature to justify the work. | Some aspects related to the methodology are included. Good justification of numerical choices backed up with the literature to some extent. | The results are good. It may not be mature and/or critical. It includes very few links to aerodynamics theories and engineering implications. Good use of the literature to strengthen the argument or discussion occasionally. | The optimization methods and results are good. It may not be mature and/or critical. It includes very few links to aerodynamics theories and engineering implications. Good use of the literature to strengthen the argument or discussion occasionally. | The discussion is good. It may not be mature and/or critical. It includes very few links to aerodynamics theories and engineering implications. Good use of the literature to strengthen the argument or discussion occasionally. | A good summary of the work done, and a few key findings are presented. A good discussion of the implications of the work. | Good use of reference. Seldom used to strengthen the argument and justify deductions. Seldom used to inform the numerical methodology and explain the flow physics (rarely). Good written communicatio n. | Good plot and Appendix quality. Not to a publication standard quality and does not adhere to the format of the whole document. May include screenshots and not be fully consistent throughout the document. Labels could be improved for clarity. Good discussion accompanies the plots. |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 45 to 49% | A satisfactory introduction. Clear and complete content. No attempt included motivate the work. Adequate aims and objectives. Adequate use of the literature to justify the work. | Missing aspects related to the methodology. Satisfactory justification of numerical choices or not backed up with the literature. | The discussion is satisfactory, but it may lack critical thinking. Clear links to the aerodynamics theories and/or engineering implications are presented. Adequate use of the literature to strengthen the argument or discussion. | Adequate summary of the work done. Adequate discussion of the results of the work. | A satisfactory introduction. Clear and complete content. No attempt included motivate the work. Adequate aims and objectives. Adequate use of the literature to justify the work. | Missing aspects related to the methodology. Satisfactory justification of numerical choices or not backed up with the literature. | The results are satisfactory, but it may lack critical thinking. Clear links to the aerodynamics theories and/or engineering implications are presented. Adequate use of the literature to strengthen the argument or discussion. | The optimization methods and results are satisfactory, but it may lack critical thinking. Clear links to the aerodynamics theories and/or engineering implications are presented. Adequate use of the literature to strengthen the argument or discussion. | The discussion is satisfactory, but it may lack critical thinking. Clear links to the aerodynamics theories and/or engineering implications are presented. Adequate use of the literature to strengthen the argument or discussion. | Adequate summary of the work done. Adequate discussion of the results of the work. | Adequate use of references. Adequate written communicatio n. Acceptable organization of the structure. | Plot and Appendix quality is acceptable. An adequate discussion that accompanies plots. |
| 40 to 44% | Relevant but poor writing introduction. No attempt to include the motivate, aim, and objectives of the work. Poor use of the literature to justify the work. | Missing aspects related to the methodology. Weak justification of numerical choices or not clearly backed up with the literature. | The discussion is weak but relevant. Links to the flow physics and/or engineering implications are weakly presented. Relevant use of the literature to strengthen the argument or discussion. | Acceptable summary of the work done. Poor but relevant discussion of the results of the work. | Relevant but poor writing introduction. No attempt to include the motivate, aim, and objectives of the work. Poor use of the literature to justify the work. | Missing aspects related to the methodology. Weak justification of numerical choices or not clearly backed up with the literature. | The results are weak but relevant. Links to the flow physics and/or engineering implications are weakly presented. Relevant use of the literature to strengthen the argument or discussion. | The optimization methods and results are weak but relevant. Links to the flow physics and/or engineering implications are weakly presented. Relevant use of the literature to strengthen the argument or discussion. | The discussion is weak but relevant. Links to the flow physics and/or engineering implications are weakly presented. Relevant use of the literature to strengthen the argument or discussion. | Acceptable summary of the work done. Poor but relevant discussion of the results of the work. | Acceptable use of references. Acceptable written communicatio n. Acceptable structure. | Poor plot and Appendix quality but acceptable. Poor and/or no discussion accompanies the plot. |
| Fail<br>30 to 39% | A weak introduction. It may be unclear and incomplete. No motivation or literature review included. | Missing aspects related to the methodology. Poor justification of numerical choices or not clearly backed up with the literature. | The discussion is poor, lacking maturity and critical thinking. No clear links to the flow physics and/or engineering implications are presented. | An incomplete summary of the work done, and no key findings of the implications of the work. | A weak introduction. It may be unclear and incomplete. No motivation or literature review included. | Missing aspects related to the methodology. Poor justification of numerical choices or not clearly backed up with the literature. | The results are poor, lacking maturity and critical thinking. No clear links to the flow physics and/or engineering implications are presented. | The optimization methods and results are poor, lacking maturity and critical thinking. No clear links to the flow physics and/or | The discussion is poor, lacking maturity and critical thinking. No clear links to the flow physics and/or engineering implications are presented. | An incomplete summary of the work done, and no key findings of the implications of the work. | Poor use of reference. No clearly used to support the argument or explain the flow physics. Poorly written communicatio n. Weak structure. | Poor plot quality and legends throughout the report. Legends may be missing or not adequate. No discussion accompanies plots. Poor |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Very poor use of the literature to strengthen the argument or discussion. | | | | Very poor use of the literature to strengthen the argument or discussion. | engineering implications are presented. Very poor use of the literature to strengthen the argument or discussion. | Very poor use of the literature to strengthen the argument or discussion. | | | quality of appendix. |
| Fail<br>01 to 29% | Extremely poor introduction. It is unclear and incomplete. No motivation, aim, and objectives or literature review included. | Extremely poor and/or irrelevant methodology. No justification for the numerical choices and/or backed up with the literature. | Extremely poor discussion. Irrelevant results. No use of the literature to strengthen the argument or discussion | Extremely poor conclusions. No discussion of the implications of the work. | Extremely poor introduction. It is unclear and incomplete. No motivation, aim, and objectives or literature review included. | Extremely poor and/or irrelevant methodology. No justification for the numerical choices and/or backed up with the literature. | Irrelevant results. No use of the literature to strengthen the argument or discussion | Irrelevant optimization methods and results. No use of the literature to strengthen the argument or discussion | Extremely poor discussion. Irrelevant results. No use of the literature to strengthen the argument or discussion | Extremely poor conclusions. No discussion of the implications of the work. | Extremely poor use of references and/or irrelevant references. Extremely poor written communicatio n. Very weak structure. | Extremely poor quality of plots along the report. No/irrelevant discussion accompanies the plots. Extremely poor use of the appendix section. |
| Fail<br>0% | The introduction is not included. | The methodology is not included for task 2, or it cannot be reproduced at all. | No results and discussion presented for task 1 | No conclusions for task 1 are presented. | The introduction is not included. | The methodology is not included for task 2 or it cannot be reproduced at all. | No Transonic results are presented for task 2 | No optimization results are presented for task 2 | No discussion presented for task 2 | No conclusions for task 2 are presented. | No references are used. | No Appendix and plots on the report are presented. |

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